

Dr.-Ing. Johannes Erfurt

AI Systems Engineer | Computer Vision Expert

github.com/JohannesErfurt | linkedin.com/in/johanneserfurt

German: Native / Muttersprache | English: Business-fluent | Russian: Very good command

PROFESSIONAL SUMMARY

AI systems engineer and computer vision specialist with a doctorate in video coding. Experience spans industrial deep-learning systems, MLOps, multi-agent applications, image and video processing, signal filtering, and international standardization.

CORE COMPETENCIES

Generative AI and multi-agent systems: LangGraph, LangChain, prompt engineering | Computer vision and machine learning: CNNs, vision transformers, fine-tuning | Software and MLOps: Python, C++, Docker, CI/CD | Image and video coding: HEVC, VVC, Wiener and learned filters

PROFESSIONAL EXPERIENCE

Software Developer for AI and Image Processing | Scansonic MI GmbH | 05/2024-Present

Develops industrial AI and computer-vision systems for automated quality monitoring of laser brazing and welding processes.

- Trains and validates defect-detection and localization models, including CNNs, ViTs, image classification, and semantic segmentation.
- Builds ML pipelines, hyperparameter-tuning workflows, data augmentation, pre-labeling infrastructure, and edge-deployment tooling.
- Reduced manual annotation effort by approximately 50% through pre-labeling infrastructure.
- Coordinates AI software releases and mentors colleagues and students.

Technologies: Python, PyTorch, Computer Vision, CNNs, Vision Transformers, MLOps

Further Education and Project Phase | DeepLearning.AI and velpTEC | 11/2023-04/2024

Intensive preparation for the transition to industrial AI development.

- Studied transformers, attention, LLMs, CNNs, RNNs, LSTMs, transfer learning, and data augmentation.

Technologies: PyTorch, TensorFlow, scikit-learn, Transformers, LLMs

World Travel and Parental Leave | Sabbatical | 11/2022-10/2023

World travel and parental leave - intercultural perspective and family focus.

Research Associate | Fraunhofer Heinrich Hertz Institute | 09/2015-10/2022

Researched image/video coding, signal filtering, learned filters, and perceptual quality assessment for HEVC and VVC.

- Developed wavelet, shearlet, Wiener-filter, classification, and artifact-reduction methods.
- Achieved more than 4% coding-efficiency improvement over VVC with a combined learned and Wiener-filter approach.
- Built reproducible C++ and Python simulation software and participated in JVET standardization.
- Contributed to WPSNR quality assessment and co-invented patent applications.

Technologies: C++, Python, MATLAB, CMake, HEVC, VVC, CNNs, CI/CD

Student Research Assistant | Fraunhofer Heinrich Hertz Institute | 08/2014-08/2015

Conducted video-compression research focused on arithmetic coding algorithms.

Technologies: Video Coding, Arithmetic Coding

Student Teaching Assistant | Technical University of Berlin | 10/2011-09/2014

Conducted mathematics and engineering tutorials and explained complex concepts to students.

EDUCATION

PhD | Technical University of Berlin | 09/2019-09/2022

Doctoral research on filter techniques in video coding, combining analytical signal processing and machine learning.

- Dissertation: Filter techniques in video coding.
- Final grade: Very good.
- Published and presented research and supervised student work.

Master of Science in Mathematics | Technische Universität Berlin | 10/2012-03/2015

Mathematics degree specializing in signal processing.

- Final grade: 1.2.
- Master's thesis: Multidimensional Inpainting, grade 1.0.

Study Abroad | Tomsk Polytechnic University, Russia | 09/2012-01/2013

International study period in Russia.

Bachelor of Science in Mathematics | Technische Universität Berlin | 10/2008-09/2012

Mathematics degree with a minor in physics.

- Final grade: 1.4.
- Bachelor's thesis: Partial Differential Equations.

Study Abroad | Durham University, England | 09/2010-06/2011

International study period in England.

SELECTED PROJECTS

MindGrow Stories | Full-Stack AI Application

A cloud-native multi-agent platform for the automated creation of personalized children's books, with modular LLM, image-generation, and VLM workflows.

Technical impact: Demonstrates end-to-end delivery of a cloud-native generative-AI product, combining multi-agent orchestration, multimodal generation, automated quality assessment, character-consistency controls, and CI/CD-based deployment.

Technologies: Next.js, FastAPI, LLMs, VLMs, Image Generation, CI/CD, Render

Wizards Story Engine | Open-Source Multi-Agent Writing System

A state-based LangGraph pipeline for planning, generating, evaluating, and iteratively revising long-form narrative text.

Technical impact: Provides an open-source reference architecture for controllable long-form generation, with specialized critic agents, iterative revision, resumable execution, human approval gates, and explicit token and cost monitoring.

Technologies: Python, LangGraph, LangChain, Pydantic, Human-in-the-Loop

<https://github.com/JohannesErfurt/wizarding-story-engine>

SELECTED PUBLICATIONS

Johannes Erfurt. "Investigation of Wiener filter techniques for in-loop filtering in video coding." Technische Universität Berlin, 2023.

DOI: <https://doi.org/10.14279/depositonce-16759>

Johannes Erfurt; Wang-Q Lim; Jonathan Pfaff; Heiko Schwarz; Detlev Marpe; Thomas Wiegand. "Shearlet-based intensity classification loop filter for video coding." Signal Processing: Image Communication, volume 115, article 116967, 2023. DOI: <https://doi.org/10.1016/j.image.2023.116967>

Wang-Q Lim; Jonathan Pfaff; Björn Stallenberger; Johannes Erfurt; Heiko Schwarz; Detlev Marpe; Thomas Wiegand. "Adaptive Loop Filter with a CNN-Based Classification." 2022 IEEE International Conference on Image Processing (ICIP), pages 1836-1840, 2022.

DOI: <https://doi.org/10.1109/ICIP46576.2022.9897666>

Johannes Erfurt; Wang-Q Lim; Heiko Schwarz; Detlev Marpe; Thomas Wiegand. "Extended multiple feature-based classifications for adaptive loop filtering." APSIPA Transactions on Signal and Information Processing, volume 8, article e28, 2019. DOI: <https://doi.org/10.1017/ATSIP.2019.19>

Johannes Erfurt; Christian R. Helmrich; Sebastian Bosse; Heiko Schwarz; Detlev Marpe; Thomas Wiegand. "A Study of the Perceptually Weighted Peak Signal-To-Noise Ratio (WPSNR) for Image Compression." 2019 IEEE International Conference on Image Processing (ICIP), pages 2339-2343, 2019. DOI: <https://doi.org/10.1109/ICIP.2019.8803307>